

Coho Salmon Project

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2026-08-28

Reading in the data

```
# List required packages
packages <- c("readxl", "boot", "tinytex")

# Identify packages that are not installed yet
new_packages <- packages[!(packages %in% installed.packages()[, "Package"])]

# Install missing packages
if (length(new_packages) > 0) {
  install.packages(new_packages)
}

# Load all packages
invisible(lapply(packages, library, character.only = TRUE))

# Ensure TinyTeX engine is installed for PDF compilation
if (!tinytex::is_tinytex()) {
  tinytex::install_tinytex()
}

# Change these if you want to run on your own machine
Saltwater <- read_excel("C:\\Users\\allen\\Downloads\\Coho data in excel (Raw).xlsx", sheet = 1)

## New names:
## * `Comments` -> `Comments...40`
## * `Comments` -> `Comments...57`

Freshwater <- read_excel("C:\\Users\\allen\\Downloads\\Coho data in excel (Raw).xlsx", sheet = 2)

## New names:
## * `Comments` -> `Comments...40`
## * `Comments` -> `Comments...57`

Freshwater$PCV1 <- as.numeric(Freshwater$PCV1)

## Warning: NAs introduced by coercion
```

```
Freshwater$PCV2 <- as.numeric(Freshwater$PCV2)
```

```
## Warning: NAs introduced by coercion
```

```
Saltwater$PCV2 <- as.numeric(Saltwater$PCV2)
```

```
## Warning: NAs introduced by coercion
```

```
Saltwater$PCV_Avg <- rowMeans(Saltwater[, c("PCV1", "PCV2")], na.rm = TRUE)
Freshwater$PCV_Avg <- rowMeans(Freshwater[, c("PCV1", "PCV2")], na.rm = TRUE)
```

```
vars <- c(
  "Heart Rate (/min)",
  "Rectal Temp \"C\"",
  "Right IR (F)",
  "Left IR (F)",
  "Dorsal IR (F)",
  "Weight (kg)",
  "Length (cm)",
  "Lac",
  "HemoCue (g/dl)",
  "PCV1",
  "PCV2",
  "TS",
  "Na (mmol/L)",
  "K (mmol/L)",
  "Cl (mmol/L)",
  "iCa (mmol/L)",
  "TCO2 (mmol/L)",
  "Glu (mg/dL)",
  "Hct (%PCV)",
  "Hb* (g/dL)",
  "BUN (mg/dL)",
  "Crea (mg/dL)",
  "WBC estimate K/ l",
  "Absolute Neutrophils K/ l",
  "Absolute immature neutrophils K/ l",
  "Absolute Lymphocytes K/ l",
  "Absolute Monocytes K/ l",
  "Absolute Eosinophils K/ l",
  "Absolute Basophils K/ l",
  "#immature RBC/100 mature RBC",
  "PCV_Avg"
)
```

```
# Forcing the columns to be numeric - normally what happens is that when you have NA in your dataset it
Saltwater[vars] <- lapply(Saltwater[vars], function(x) {
```

```
  x <- trimws(x) # remove extra spaces
  x[x %in% c("NA", "****", "")] <- NA

  as.numeric(x)
```

```
})
```

```
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
```

```
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
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## Warning in FUN(X[[i]], ...): NAs introduced by coercion
```

```
Freshwater[vars] <- lapply(Freshwater[vars], function(x) {

  x <- trimws(x)                # remove extra spaces
  x[x %in% c("NA", "***", "")] <- NA

  as.numeric(x)

})
```

```
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
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```

Analyzing Saltwater Salmon

Pages 8 - 10

1. Determine if data is Gaussian/Nongaussian based on Shapiro Wilk
2. Remove outliers
3. Calculate reference Intervals

If sample size below 20: Do not need to calculate RI.

NEED:

- Histogram of Data (Including outliers)
- Mean (if Gaussian) or Median (if not Gaussian)
- Minimum and Maximum Values

Gaussian? (Shapiro Wilk)

```

shapiro_p <- numeric(length(vars))
n_values <- numeric(length(vars))

names(shapiro_p) <- vars
names(n_values) <- vars

shapiro_p <- numeric(length(vars))
shapiro_W <- numeric(length(vars))
n_values <- numeric(length(vars))

names(shapiro_p) <- vars
names(shapiro_W) <- vars
names(n_values) <- vars

for (v in vars) {

  x <- na.omit(Saltwater[[v]])
  n_values[v] <- length(x)

  if (length(x) >= 3 && length(unique(x)) > 1) {

    test <- shapiro.test(x)
    shapiro_p[v] <- test$p.value
    shapiro_W[v] <- test$statistic

  } else {

    shapiro_p[v] <- NA
    shapiro_W[v] <- NA

  }
}

Saltwater_Normality <- data.frame(
  Variable = vars,
  N = n_values,
  Shapiro_p_value = shapiro_p,
  Normal = ifelse(shapiro_p > 0.05, "Yes", "No")
)

```

```
Saltwater_Normality
```

##	Variable	N
## Heart Rate (/min)	Heart Rate (/min)	25
## Rectal Temp "C"	Rectal Temp "C"	25
## Right IR (F)	Right IR (F)	25
## Left IR (F)	Left IR (F)	25
## Dorsal IR (F)	Dorsal IR (F)	25
## Weight (kg)	Weight (kg)	25
## Length (cm)	Length (cm)	25
## Lac	Lac	25
## HemoCue (g/dl)	HemoCue (g/dl)	25
## PCV1	PCV1	25

## PCV2		PCV2	24
## TS		TS	25
## Na (mmol/L)		Na (mmol/L)	25
## K (mmol/L)		K (mmol/L)	22
## Cl (mmol/L)		Cl (mmol/L)	6
## iCa (mmol/L)		iCa (mmol/L)	25
## TC02 (mmol/L)		TC02 (mmol/L)	25
## Glu (mg/dL)		Glu (mg/dL)	25
## Hct (%PCV)		Hct (%PCV)	25
## Hb* (g/dL)		Hb* (g/dL)	25
## BUN (mg/dL)		BUN (mg/dL)	0
## Crea (mg/dL)		Crea (mg/dL)	1
## WBC estimate K/ l		WBC estimate K/ l	15
## Absolute Neutrophils K/ l		Absolute Neutrophils K/ l	15
## Absolute immature neutrophils K/ l	Absolute	immature neutrophils K/ l	15
## Absolute Lymphocytes K/ l		Absolute Lymphocytes K/ l	15
## Absolute Monocytes K/ l		Absolute Monocytes K/ l	15
## Absolute Eosinophils K/ l		Absolute Eosinophils K/ l	15
## Absolute Basophils K/ l		Absolute Basophils K/ l	15
## #immature RBC/100 mature RBC		#immature RBC/100 mature RBC	24
## PCV_Avg		PCV_Avg	25
##	Shapiro_p_value	Normal	
## Heart Rate (/min)	1.553174e-06	No	
## Rectal Temp "C"	1.271147e-03	No	
## Right IR (F)	3.502422e-02	No	
## Left IR (F)	1.031634e-01	Yes	
## Dorsal IR (F)	3.293985e-02	No	
## Weight (kg)	6.152901e-02	Yes	
## Length (cm)	4.898003e-01	Yes	
## Lac	6.722248e-01	Yes	
## HemoCue (g/dl)	1.471648e-01	Yes	
## PCV1	4.605068e-01	Yes	
## PCV2	2.011202e-01	Yes	
## TS	1.821697e-01	Yes	
## Na (mmol/L)	2.240122e-01	Yes	
## K (mmol/L)	6.934870e-01	Yes	
## Cl (mmol/L)	9.605550e-01	Yes	
## iCa (mmol/L)	7.183509e-01	Yes	
## TC02 (mmol/L)	1.917804e-01	Yes	
## Glu (mg/dL)	3.682322e-01	Yes	
## Hct (%PCV)	5.485965e-04	No	
## Hb* (g/dL)	2.232670e-03	No	
## BUN (mg/dL)	NA	<NA>	
## Crea (mg/dL)	NA	<NA>	
## WBC estimate K/ l	9.512041e-02	Yes	
## Absolute Neutrophils K/ l	2.646226e-02	No	
## Absolute immature neutrophils K/ l	5.979596e-03	No	
## Absolute Lymphocytes K/ l	5.884916e-01	Yes	
## Absolute Monocytes K/ l	5.862718e-02	Yes	
## Absolute Eosinophils K/ l	NA	<NA>	
## Absolute Basophils K/ l	NA	<NA>	
## #immature RBC/100 mature RBC	8.398320e-01	Yes	
## PCV_Avg	3.826578e-01	Yes	

Gaussian Data - Removing outliers

Page 8 of the Guidelines for RI from ASVCP (Horn's Algorithm)

Page 9: Eliminate outliers, retest, eliminate again.

Directly from the guidelines: ***NOTE: The RI study summary document should reflect the number of outliers eliminated, the method used to identify them and the final number (n) from which RI are determined.***

```
normal_vars <- Saltwater_Normality$Variable[
  !is.na(Saltwater_Normality$Normal) &
  Saltwater_Normality$Normal == "Yes"
]

Saltwater_NoOutliers <- Saltwater

outliers_removed_salt <- numeric(length(normal_vars))
final_nonNA_salt <- numeric(length(normal_vars))

names(outliers_removed_salt) <- normal_vars
names(final_nonNA_salt) <- normal_vars

for (v in normal_vars) {

  removed_total <- 0

  repeat {

    x <- Saltwater_NoOutliers[[v]]
    x_noNA <- na.omit(x)

    Q1 <- quantile(x_noNA, 0.25)
    Q3 <- quantile(x_noNA, 0.75)
    IQR_value <- IQR(x_noNA)

    lower <- Q1 - 1.5 * IQR_value
    upper <- Q3 + 1.5 * IQR_value

    outliers <- which(x < lower | x > upper)

    if (length(outliers) == 0) break

    removed_total <- removed_total + length(outliers)

    Saltwater_NoOutliers[[v]][outliers] <- NA

  }

  outliers_removed_salt[v] <- removed_total
  final_nonNA_salt[v] <- sum(!is.na(Saltwater_NoOutliers[[v]]))
}

Saltwater_results <- data.frame(
  Variable = normal_vars,
```

```

    Outliers_Removed = outliers_removed_salt,
    Final_NonNA = final_nonNA_salt
)

rownames(Saltwater_results) <- NULL
Saltwater_results

```

```

##           Variable Outliers_Removed Final_NonNA
## 1           Left IR (F)                3         22
## 2           Weight (kg)                3         22
## 3           Length (cm)                0         25
## 4              Lac                   0         25
## 5          HemoCue (g/dl)              4         21
## 6              PCV1                   0         25
## 7              PCV2                   0         24
## 8              TS                     2         23
## 9              Na (mmol/L)             3         22
## 10             K (mmol/L)              0         22
## 11             Cl (mmol/L)             0          6
## 12             iCa (mmol/L)            4         21
## 13            TCO2 (mmol/L)            1         24
## 14             Glu (mg/dL)             0         25
## 15          WBC estimate K/ l           3         12
## 16 Absolute Lymphocytes K/ l           0         15
## 17 Absolute Monocytes K/ l            0         15
## 18 #immature RBC/100 mature RBC        0         24
## 19              PCV_Avg               0         25

```

Mean $\pm$ 1.645 * SD for $20 \leq n < 40$ and Gaussian

```

# qnorm(0.95) Number to multiply the SD by for Gaussian Data
# Uncomment this install.packages below if you need the package to run
# install.packages("boot")
library(boot)
set.seed(352026)

# Function used by bootstrap
bootstrap_RI <- function(data, indices) {

  d <- data[indices]

  mean_d <- mean(d)
  sd_d <- sd(d)

  c(
    mean_d - 1.645 * sd_d,
    mean_d + 1.645 * sd_d
  )
}

vars_param <- Saltwater_results$Variable[

```

```

Saltwater_results$Final_NonNA >= 20 &
Saltwater_results$Final_NonNA < 40
]

param_results <- data.frame()

for (v in vars_param) {

  x <- na.omit(Saltwater_NoOutliers[[v]])

  n <- length(x)
  mean_x <- mean(x)
  sd_x <- sd(x)
  min_x <- min(x)
  max_x <- max(x)

  RI_lower <- mean_x - 1.645 * sd_x
  RI_upper <- mean_x + 1.645 * sd_x

  # Bootstrap CI
  results <- boot(x, statistic = bootstrap_RI, R = 10000)

  lower_ci <- boot.ci(results, type = "perc", index = 1)$percent[4:5]
  upper_ci <- boot.ci(results, type = "perc", index = 2)$percent[4:5]

  param_results <- rbind(
    param_results,
    data.frame(
      Variable = v,
      N = n,
      OutliersRemoved = Saltwater_results$Outliers_Removed[Saltwater_results$Variable == v],
      Minimum = min_x,
      Maximum = max_x,
      Mean = mean_x,
      SD = sd_x,
      RI_lower = RI_lower,
      RI_upper = RI_upper,
      CI90_lower_L = lower_ci[1],
      CI90_lower_U = lower_ci[2],
      CI90_upper_L = upper_ci[1],
      CI90_upper_U = upper_ci[2]
    )
  )
}

rownames(param_results) <- NULL
param_results

```

```

##           Variable  N OutliersRemoved Minimum Maximum      Mean
## 1      Left IR (F) 22              3    49.10     58.1  53.640909
## 2      Weight (kg) 22              3     1.50      3.8   2.600909
## 3      Length (cm) 25              0    56.00    71.5  63.840000
## 4              Lac 25              0     0.80     6.2   2.976000

```



```
## 5          HemoCue (g/dl) 21          4    7.80    11.4    9.642857
## 6              PCV1 25          0   32.00    56.0   42.920000
## 7              PCV2 24          0   32.00    55.0   43.458333
## 8              TS 23          2    5.20    10.2    7.521739
## 9              Na (mmol/L) 22        3  148.00   157.0  151.772727
## 10             K (mmol/L) 22          0    2.30     5.7    4.245455
## 11             iCa (mmol/L) 21         4    1.32     1.7    1.518571
## 12             TC02 (mmol/L) 24        1    6.00    13.0    8.958333
## 13             Glu (mg/dL) 25          0   52.00    78.0   65.440000
## 14 #immature RBC/100 mature RBC 24      0    7.00    52.0   28.875000
## 15              PCV_Avg 25          0   32.00    55.5   43.200000
##          SD      RI_lower  RI_upper  CI90_lower_L  CI90_lower_U  CI90_upper_L
## 1  2.36877688  49.7442711  57.537547  48.603767  51.185456  55.909218
## 2  0.59045510  1.6296104  3.572208  1.357443  1.968882  3.136592
## 3  4.20743390  56.9187712  70.761229  54.912874  59.514285  68.323233
## 4  1.21835955  0.9717985  4.980201  0.407699  1.669265  4.098273
## 5  0.85297798  8.2397084  11.046006  7.713979  8.877884  10.500974
## 6  6.48022633  32.2600277  53.579972  29.596038  35.634386  49.200332
## 7  6.66580610  32.4930823  54.423584  29.652744  36.076266  49.775430
## 8  1.20074088  5.5465204  9.496958  4.941248  6.336614  8.741671
## 9  2.15874342  148.2215943  155.323860  147.201035  149.520947  153.701580
## 10 0.91063937  2.7474528  5.743456  2.219613  3.393573  5.224207
## 11 0.09986419  1.3542948  1.682848  1.292682  1.434356  1.620110
## 12 1.85282784  5.9104315  12.006235  5.127747  6.932339  10.669084
## 13 7.40540343  53.2581114  77.621889  50.093294  57.252335  72.765805
## 14 11.30337579  10.2809468  47.469053  4.320034  17.707924  40.940338
## 15 6.49037749  32.5233290  53.876671  29.760057  35.899027  49.446533
##      CI90_upper_U
## 1      58.814982
## 2       3.935151
## 3      72.655063
## 4       5.798626
## 5      11.497856
## 6      57.074160
## 7      57.731528
## 8      10.161778
## 9     156.738947
## 10      6.142946
## 11      1.732051
## 12     13.085827
## 13     81.189199
## 14     52.988397
## 15     57.103928
```

Finishing off the Gaussian data (n below 20)

Need:

- Histogram of data
- Mean
- Minimum and Maximum values

```
vars_hist <- Saltwater_results$Variable[
  Saltwater_results$Final_NonNA < 20
```

```

]

results_summary_salt <- data.frame()

for (v in vars_hist) {

  x <- na.omit(Saltwater_NoOutliers[[v]])

  hist(x,
        main = paste("Histogram of", v, "(Saltwater - Gaussian)"),
        xlab = v,
        col = "lightblue",
        border = "black")
  cat("\n\\newpage\n")

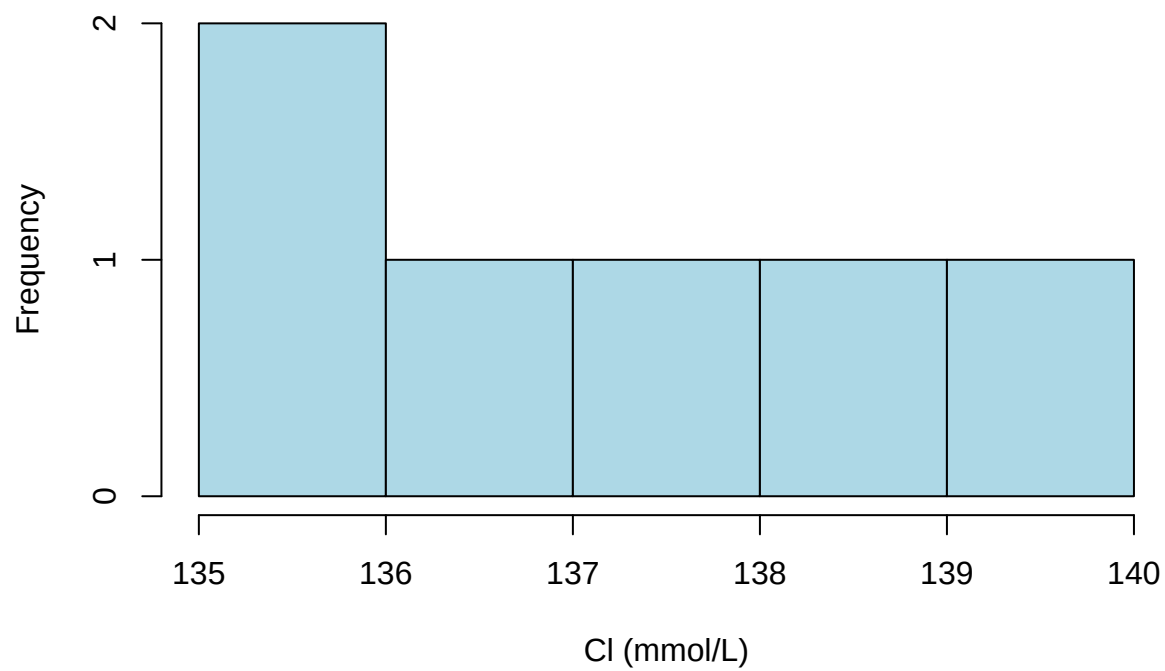
  n <- length(x)
  mean_val <- mean(x)
  min_val <- min(x)
  max_val <- max(x)

  values_sorted <- paste(sort(x), collapse = ", ")

  results_summary_salt <- rbind(
    results_summary_salt,
    data.frame(
      Variable = v,
      N = n,
      OutliersRemoved = Saltwater_results$Outliers_Removed[Saltwater_results$Variable == v],
      Mean = mean_val,
      Minimum = min_val,
      Maximum = max_val,
      Values = values_sorted
    )
  )
}

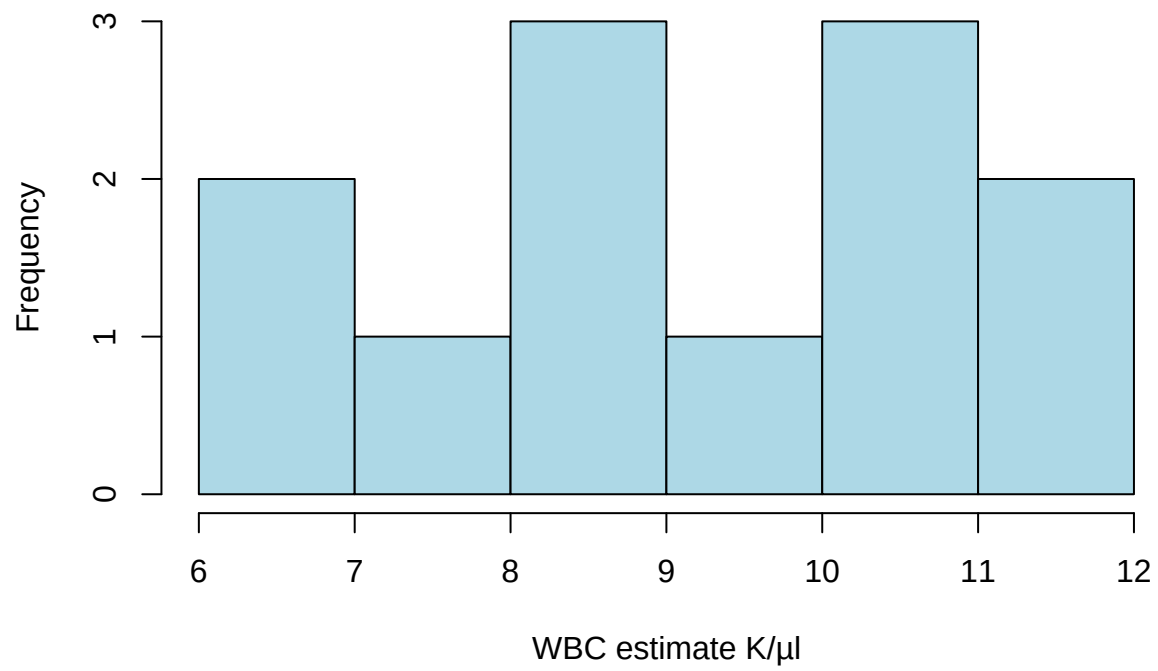
```

Histogram of Cl (mmol/L) (Saltwater – Gaussian)



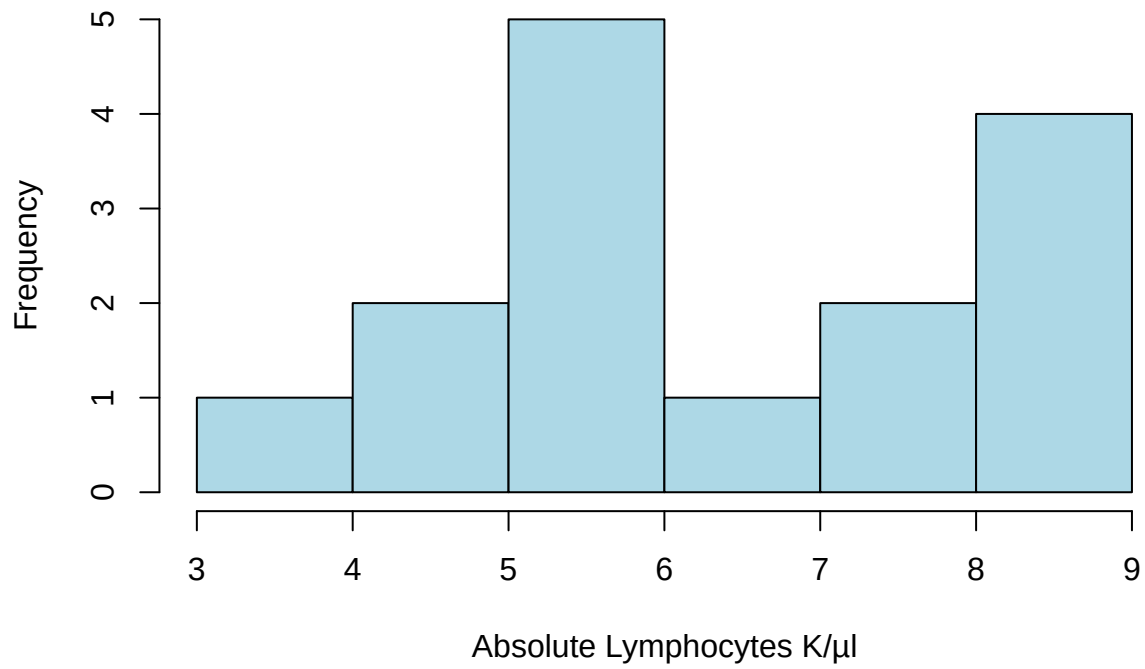
\newpage

Histogram of WBC estimate K/ μ l (Saltwater – Gaussian)



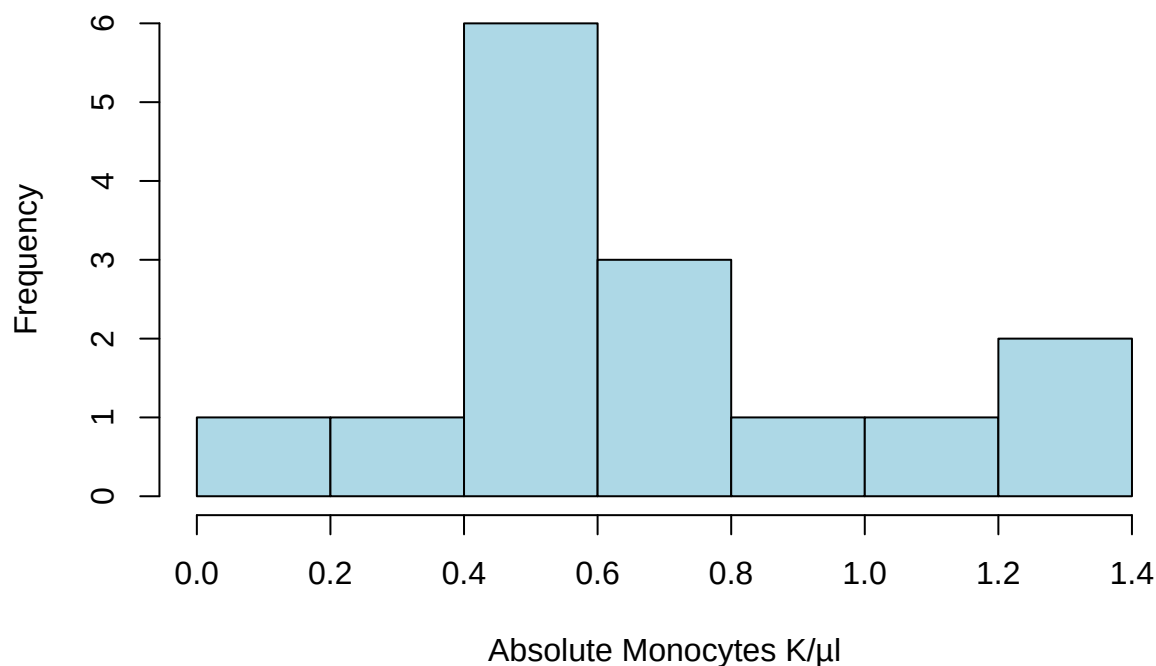
\newpage

Histogram of Absolute Lymphocytes K/ μ l (Saltwater – Gaussian)



\newpage

Histogram of Absolute Monocytes K/ μ l (Saltwater – Gaussian)



```
##
## \newpage
```

```
results_summary_salt
```

```
##           Variable  N OutliersRemoved      Mean Minimum Maximum
## 1           Cl (mmol/L) 6              0 137.5000000 135.00 140.0
## 2      WBC estimate K/ l 12              3  9.2416667   6.10  11.7
## 3 Absolute Lymphocytes K/ l 15              0  6.3666667   3.00   8.8
## 4 Absolute Monocytes K/ l 15              0  0.6986667   0.15   1.4
##
##                                     Values
## 1                                     135, 136, 137, 138, 139, 140
## 2              6.1, 6.9, 7.5, 8.1, 8.8, 8.9, 9.8, 10.2, 10.6, 10.6, 11.7, 11.7
## 3              3, 4.4, 4.5, 5.2, 5.4, 5.7, 5.8, 5.9, 6.7, 7.2, 7.6, 8.2, 8.3, 8.8, 8.8
## 4 0.15, 0.32, 0.47, 0.48, 0.49, 0.49, 0.51, 0.53, 0.62, 0.7, 0.78, 0.94, 1.2, 1.4, 1.4
```

Saltwater not Gaussian $20 \leq n < 40$

```
library(boot)
set.seed(352026)

bootstrap_RI_np <- function(data, indices) {
```

```

d <- data[indices]

c(
  quantile(d, 0.05),
  quantile(d, 0.95)
)
}

nonnormal_vars <- Saltwater_Normality$Variable[
  !is.na(Saltwater_Normality$Normal) &
  Saltwater_Normality$Normal == "No"
]

vars_np <- nonnormal_vars[
  sapply(nonnormal_vars, function(v)
    sum(!is.na(Saltwater_NoOutliers[[v]])) >= 20 &
    sum(!is.na(Saltwater_NoOutliers[[v]])) < 40
  )
]

nonnormal_results <- data.frame()

for (v in vars_np) {

  x <- na.omit(Saltwater_NoOutliers[[v]])

  n <- length(x)
  min_x <- min(x)
  max_x <- max(x)
  med_x <- median(x)

  RI_lower <- quantile(x, 0.05)
  RI_upper <- quantile(x, 0.95)

  # Bootstrap
  results <- boot(x, statistic = bootstrap_RI_np, R = 10000)

  lower_ci <- boot.ci(results, type = "perc", index = 1)$percent[4:5]
  upper_ci <- boot.ci(results, type = "perc", index = 2)$percent[4:5]

  nonnormal_results <- rbind(
    nonnormal_results,
    data.frame(
      Variable = v,
      N = n,
      Minimum = min_x,
      Maximum = max_x,
      Median = med_x,
      RI_lower = RI_lower,
      RI_upper = RI_upper,
      CI90_lower_L = lower_ci[1],
      CI90_lower_U = lower_ci[2],
      CI90_upper_L = upper_ci[1],

```

```

        CI90_upper_U = upper_ci[2]
    )
}

rownames(nonnormal_results) <- NULL
nonnormal_results

```

```

##           Variable  N Minimum Maximum Median RI_lower RI_upper CI90_lower_L
## 1 Heart Rate (/min) 25     9.0    56.0   18.0     9.00    26.40         9.0
## 2 Rectal Temp "C"  25    11.0    15.5   12.1    11.14    13.38        11.0
## 3 Right IR (F)    25    50.2    64.9   54.5    50.64    60.28        50.2
## 4 Dorsal IR (F)   25    49.8    63.3   53.8    50.10    58.94        49.8
## 5 Hct (%PCV)     25    20.0    47.0   26.0    21.00    32.00        20.0
## 6 Hb* (g/dL)     25     6.8    16.0    8.8     7.10    11.70         6.8
##  CI90_lower_U CI90_upper_L CI90_upper_U
## 1         12.00         19.60         56.0
## 2         11.32         12.58         15.5
## 3         52.54         56.30         64.9
## 4         52.00         55.60         63.3
## 5         24.20         30.00         47.0
## 6          8.20         10.20         16.0

```

Saltwater not Gaussian (n below 20)

```

nonnormal_vars <- Saltwater_Normality$Variable[
  !is.na(Saltwater_Normality$Normal) &
  Saltwater_Normality$Normal == "No"
]

vars_small_np <- nonnormal_vars[
  sapply(nonnormal_vars, function(v)
    sum(!is.na(Saltwater[[v]])) < 20
  )
]

results_summary_small_np <- data.frame()

for (v in vars_small_np) {
  x <- na.omit(Saltwater[[v]])

  hist(x,
    main = paste("Histogram of", v, "\n(Saltwater - Not Gaussian)",
    xlab = v,
    col = "lightblue",
    border = "black")

  cat("\n\\newpage\n")

```



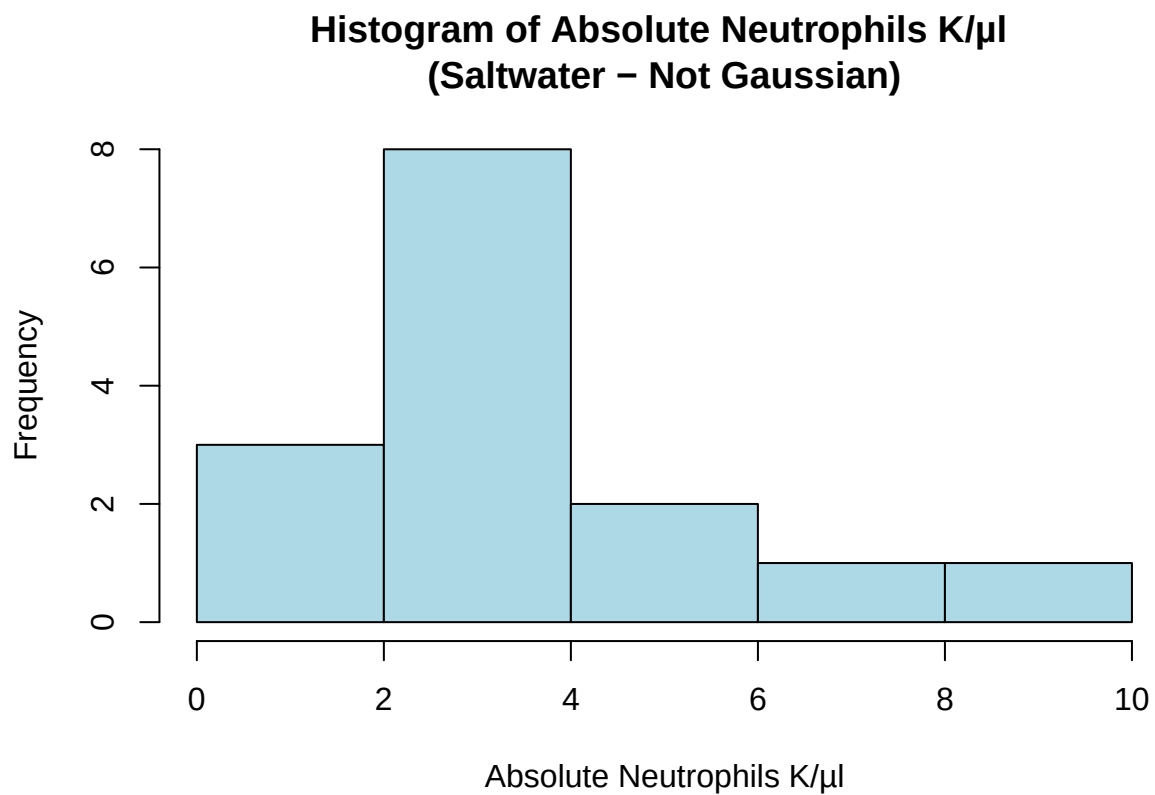
```

n <- length(x)
median_val <- median(x)
min_val <- min(x)
max_val <- max(x)

values_sorted <- paste(sort(x), collapse = ", ")

results_summary_small_np <- rbind(
  results_summary_small_np,
  data.frame(
    Variable = v,
    N = n,
    Median = median_val,
    Minimum = min_val,
    Maximum = max_val,
    Values = values_sorted
  )
)
}

```

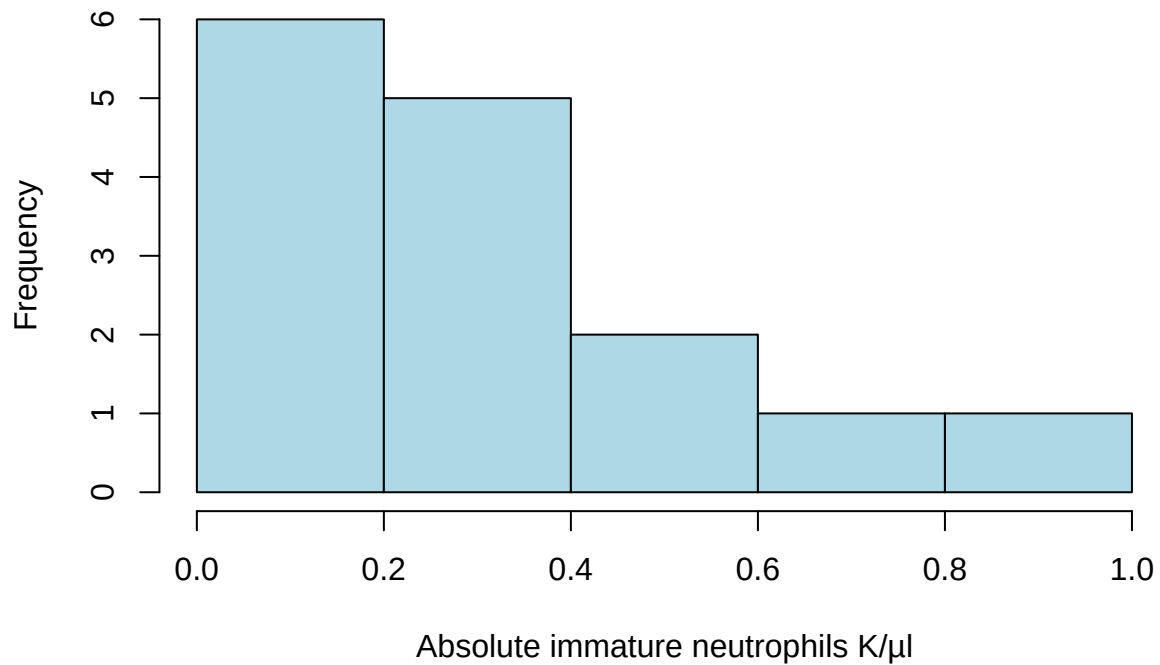


```

##
## \newpage

```

**Histogram of Absolute immature neutrophils K/ μ l
(Saltwater – Not Gaussian)**



```
##
## \newpage
```

```
results_summary_small_np
```

```
##
##          Variable  N Median Minimum Maximum
## 1 Absolute Neutrophils K/ l 15  2.80    1.10    8.40
## 2 Absolute immature neutrophils K/ l 15  0.23    0.09    0.84
##
##                                     Values
## 1          1.1, 1.5, 1.8, 2.1, 2.5, 2.8, 2.8, 2.8, 2.9, 3.4, 3.8, 4.3, 4.9, 7.2, 8.4
## 2 0.09, 0.11, 0.12, 0.14, 0.15, 0.2, 0.21, 0.23, 0.24, 0.26, 0.35, 0.49, 0.57, 0.8, 0.84
```

Saltwater Results

```
param_results
```

```
##
##          Variable  N OutliersRemoved Minimum Maximum      Mean
## 1 Left IR (F) 22          3    49.10    58.1  53.640909
## 2 Weight (kg) 22          3     1.50     3.8   2.600909
## 3 Length (cm) 25          0    56.00    71.5  63.840000
## 4 Lac 25          0     0.80     6.2   2.976000
## 5 HemoCue (g/dl) 21          4     7.80    11.4   9.642857
```

```

## 6          PCV1 25          0 32.00 56.0 42.920000
## 7          PCV2 24          0 32.00 55.0 43.458333
## 8          TS 23           2 5.20 10.2 7.521739
## 9          Na (mmol/L) 22      3 148.00 157.0 151.772727
## 10         K (mmol/L) 22       0 2.30 5.7 4.245455
## 11         iCa (mmol/L) 21      4 1.32 1.7 1.518571
## 12         TC02 (mmol/L) 24     1 6.00 13.0 8.958333
## 13         Glu (mg/dL) 25       0 52.00 78.0 65.440000
## 14 #immature RBC/100 mature RBC 24 0 7.00 52.0 28.875000
## 15          PCV_Avg 25          0 32.00 55.5 43.200000
##          SD      RI_lower  RI_upper CI90_lower_L CI90_lower_U CI90_upper_L
## 1  2.36877688 49.7442711 57.537547 48.603767 51.185456 55.909218
## 2  0.59045510 1.6296104 3.572208 1.357443 1.968882 3.136592
## 3  4.20743390 56.9187712 70.761229 54.912874 59.514285 68.323233
## 4  1.21835955 0.9717985 4.980201 0.407699 1.669265 4.098273
## 5  0.85297798 8.2397084 11.046006 7.713979 8.877884 10.500974
## 6  6.48022633 32.2600277 53.579972 29.596038 35.634386 49.200332
## 7  6.66580610 32.4930823 54.423584 29.652744 36.076266 49.775430
## 8  1.20074088 5.5465204 9.496958 4.941248 6.336614 8.741671
## 9  2.15874342 148.2215943 155.323860 147.201035 149.520947 153.701580
## 10 0.91063937 2.7474528 5.743456 2.219613 3.393573 5.224207
## 11 0.09986419 1.3542948 1.682848 1.292682 1.434356 1.620110
## 12 1.85282784 5.9104315 12.006235 5.127747 6.932339 10.669084
## 13 7.40540343 53.2581114 77.621889 50.093294 57.252335 72.765805
## 14 11.30337579 10.2809468 47.469053 4.320034 17.707924 40.940338
## 15 6.49037749 32.5233290 53.876671 29.760057 35.899027 49.446533
##          CI90_upper_U
## 1 58.814982
## 2 3.935151
## 3 72.655063
## 4 5.798626
## 5 11.497856
## 6 57.074160
## 7 57.731528
## 8 10.161778
## 9 156.738947
## 10 6.142946
## 11 1.732051
## 12 13.085827
## 13 81.189199
## 14 52.988397
## 15 57.103928

```

results_summary_salt

```

##          Variable  N OutliersRemoved      Mean Minimum Maximum
## 1          Cl (mmol/L) 6          0 137.5000000 135.00 140.0
## 2      WBC estimate K/ l 12          3 9.2416667 6.10 11.7
## 3 Absolute Lymphocytes K/ l 15          0 6.3666667 3.00 8.8
## 4 Absolute Monocytes K/ l 15          0 0.6986667 0.15 1.4
##
##                                     Values
## 1                                     135, 136, 137, 138, 139, 140
## 2          6.1, 6.9, 7.5, 8.1, 8.8, 8.9, 9.8, 10.2, 10.6, 10.6, 11.7, 11.7
## 3          3, 4.4, 4.5, 5.2, 5.4, 5.7, 5.8, 5.9, 6.7, 7.2, 7.6, 8.2, 8.3, 8.8, 8.8

```

```
## 4 0.15, 0.32, 0.47, 0.48, 0.49, 0.49, 0.51, 0.53, 0.62, 0.7, 0.78, 0.94, 1.2, 1.4, 1.4
```

nonnormal_results

```
##          Variable  N Minimum Maximum Median RI_lower RI_upper CI90_lower_L
## 1 Heart Rate (/min) 25    9.0    56.0   18.0    9.00    26.40         9.0
## 2 Rectal Temp "C" 25   11.0    15.5   12.1   11.14    13.38        11.0
## 3 Right IR (F) 25   50.2    64.9   54.5   50.64    60.28        50.2
## 4 Dorsal IR (F) 25   49.8    63.3   53.8   50.10    58.94        49.8
## 5 Hct (%PCV) 25   20.0    47.0   26.0   21.00    32.00        20.0
## 6 Hb* (g/dL) 25    6.8    16.0    8.8    7.10    11.70         6.8
## CI90_lower_U CI90_upper_L CI90_upper_U
## 1      12.00      19.60      56.0
## 2      11.32      12.58      15.5
## 3      52.54      56.30      64.9
## 4      52.00      55.60      63.3
## 5      24.20      30.00      47.0
## 6       8.20      10.20      16.0
```

results_summary_small_np

```
##          Variable  N Median Minimum Maximum
## 1 Absolute Neutrophils K/ l 15  2.80    1.10    8.40
## 2 Absolute immature neutrophils K/ l 15  0.23    0.09    0.84
##
## 1          1.1, 1.5, 1.8, 2.1, 2.5, 2.8, 2.8, 2.8, 2.9, 3.4, 3.8, 4.3, 4.9, 7.2, 8.4
## 2 0.09, 0.11, 0.12, 0.14, 0.15, 0.2, 0.21, 0.23, 0.24, 0.26, 0.35, 0.49, 0.57, 0.8, 0.84
```

No category

```
Saltwater_Normality$Variable[is.na(Saltwater_Normality$Normal)]
```

```
## [1] "BUN (mg/dL)" "Crea (mg/dL)"
## [3] "Absolute Eosinophils K/ l" "Absolute Basophils K/ l"
```

Analyzing Freshwater Salmon

Gaussian? (Shapiro - Wilk)

```
shapiro_p <- numeric(length(vars))
shapiro_W <- numeric(length(vars))
n_values <- numeric(length(vars))

names(shapiro_p) <- vars
names(shapiro_W) <- vars
names(n_values) <- vars

for (v in vars) {
  x <- na.omit(Freshwater[[v]])
```

```

n_values[v] <- length(x)

if (length(x) >= 3 && length(unique(x)) > 1) {

  test <- shapiro.test(x)
  shapiro_p[v] <- test$p.value
  shapiro_W[v] <- test$statistic

} else {

  shapiro_p[v] <- NA
  shapiro_W[v] <- NA

}
}

Freshwater_Normality <- data.frame(
  Variable = vars,
  N = n_values,
  Shapiro_p_value = shapiro_p,
  Normal = ifelse(shapiro_p > 0.05, "Yes", "No")
)

Freshwater_Normality

```

##	Variable	N
## Heart Rate (/min)	Heart Rate (/min)	21
## Rectal Temp "C"	Rectal Temp "C"	21
## Right IR (F)	Right IR (F)	21
## Left IR (F)	Left IR (F)	21
## Dorsal IR (F)	Dorsal IR (F)	21
## Weight (kg)	Weight (kg)	7
## Length (cm)	Length (cm)	7
## Lac	Lac	21
## HemoCue (g/dl)	HemoCue (g/dl)	21
## PCV1	PCV1	20
## PCV2	PCV2	20
## TS	TS	18
## Na (mmol/L)	Na (mmol/L)	21
## K (mmol/L)	K (mmol/L)	2
## Cl (mmol/L)	Cl (mmol/L)	18
## iCa (mmol/L)	iCa (mmol/L)	21
## TC02 (mmol/L)	TC02 (mmol/L)	20
## Glu (mg/dL)	Glu (mg/dL)	20
## Hct (%PCV)	Hct (%PCV)	21
## Hb* (g/dL)	Hb* (g/dL)	21
## BUN (mg/dL)	BUN (mg/dL)	1
## Crea (mg/dL)	Crea (mg/dL)	0
## WBC estimate K/ l	WBC estimate K/ l	21
## Absolute Neutrophils K/ l	Absolute Neutrophils K/ l	21
## Absolute immature neutrophils K/ l	Absolute immature neutrophils K/ l	21
## Absolute Lymphocytes K/ l	Absolute Lymphocytes K/ l	21
## Absolute Monocytes K/ l	Absolute Monocytes K/ l	21

## Absolute Eosinophils K/ l	Absolute Eosinophils K/ l	21
## Absolute Basophils K/ l	Absolute Basophils K/ l	21
## #immature RBC/100 mature RBC	#immature RBC/100 mature RBC	21
## PCV_Avg	PCV_Avg	21
##	Shapiro_p_value	Normal
## Heart Rate (/min)	0.018244536	No
## Rectal Temp "C"	0.413128447	Yes
## Right IR (F)	0.899528230	Yes
## Left IR (F)	0.269582035	Yes
## Dorsal IR (F)	0.174882235	Yes
## Weight (kg)	0.812255574	Yes
## Length (cm)	0.203213724	Yes
## Lac	0.137921402	Yes
## HemoCue (g/dl)	0.466069661	Yes
## PCV1	0.099233407	Yes
## PCV2	0.707013702	Yes
## TS	0.353367494	Yes
## Na (mmol/L)	0.054359606	Yes
## K (mmol/L)	NA	<NA>
## Cl (mmol/L)	0.023301650	No
## iCa (mmol/L)	0.003006663	No
## TC02 (mmol/L)	0.250112149	Yes
## Glu (mg/dL)	0.106588294	Yes
## Hct (%PCV)	0.632128796	Yes
## Hb* (g/dL)	0.622357416	Yes
## BUN (mg/dL)	NA	<NA>
## Crea (mg/dL)	NA	<NA>
## WBC estimate K/ l	0.882622983	Yes
## Absolute Neutrophils K/ l	0.133593603	Yes
## Absolute immature neutrophils K/ l	0.243310702	Yes
## Absolute Lymphocytes K/ l	0.381568034	Yes
## Absolute Monocytes K/ l	0.432382751	Yes
## Absolute Eosinophils K/ l	NA	<NA>
## Absolute Basophils K/ l	NA	<NA>
## #immature RBC/100 mature RBC	0.006606397	No
## PCV_Avg	0.638202432	Yes

Gaussian Data - Removing outliers

```
normal_vars_fw <- Freshwater_Normality$Variable[
  !is.na(Freshwater_Normality$Normal) &
  Freshwater_Normality$Normal == "Yes"
]

Freshwater_NoOutliers <- Freshwater

outliers_removed_fw <- numeric(length(normal_vars_fw))
final_nonNA_fw <- numeric(length(normal_vars_fw))

names(outliers_removed_fw) <- normal_vars_fw
names(final_nonNA_fw) <- normal_vars_fw
```

```

for (v in normal_vars_fw) {

  removed_total <- 0

  repeat {

    x <- Freshwater_NoOutliers[[v]]
    x_noNA <- na.omit(x)

    Q1 <- quantile(x_noNA, 0.25)
    Q3 <- quantile(x_noNA, 0.75)
    IQR_value <- IQR(x_noNA)

    lower <- Q1 - 1.5 * IQR_value
    upper <- Q3 + 1.5 * IQR_value

    outliers <- which(x < lower | x > upper)

    if (length(outliers) == 0) break

    removed_total <- removed_total + length(outliers)

    Freshwater_NoOutliers[[v]][outliers] <- NA

  }

  outliers_removed_fw[v] <- removed_total
  final_nonNA_fw[v] <- sum(!is.na(Freshwater_NoOutliers[[v]]))
}

Freshwater_results <- data.frame(
  Variable = normal_vars_fw,
  Outliers_Removed = outliers_removed_fw,
  Final_NonNA = final_nonNA_fw
)

rownames(Freshwater_results) <- NULL
Freshwater_results

```

##	Variable	Outliers_Removed	Final_NonNA
## 1	Rectal Temp "C"	1	20
## 2	Right IR (F)	0	21
## 3	Left IR (F)	0	21
## 4	Dorsal IR (F)	0	21
## 5	Weight (kg)	0	7
## 6	Length (cm)	0	7
## 7	Lac	1	20
## 8	HemoCue (g/dl)	0	21
## 9	PCV1	0	20
## 10	PCV2	0	20
## 11	TS	2	16
## 12	Na (mmol/L)	0	21
## 13	TCO2 (mmol/L)	3	17

## 14	Glu (mg/dL)	1	19
## 15	Hct (%PCV)	0	21
## 16	Hb* (g/dL)	0	21
## 17	WBC estimate K/ l	0	21
## 18	Absolute Neutrophils K/ l	0	21
## 19	Absolute immature neutrophils K/ l	0	21
## 20	Absolute Lymphocytes K/ l	0	21
## 21	Absolute Monocytes K/ l	2	19
## 22	PCV_Avg	0	21

Mean $\pm$ 1.645 * SD for $20 \leq n < 40$ and Gaussian

```
library(boot)
set.seed(352026)

bootstrap_RI <- function(data, indices) {

  d <- data[indices]

  mean_d <- mean(d)
  sd_d <- sd(d)

  c(
    mean_d - 1.645 * sd_d,
    mean_d + 1.645 * sd_d
  )
}

vars_param_fw <- Freshwater_results$Variable[
  Freshwater_results$Final_NonNA >= 20 &
  Freshwater_results$Final_NonNA < 40
]

param_results_fw <- data.frame()

for (v in vars_param_fw) {

  x <- na.omit(Freshwater_NoOutliers[[v]])

  n <- length(x)
  mean_x <- mean(x)
  sd_x <- sd(x)
  min_x <- min(x)
  max_x <- max(x)

  RI_lower <- mean_x - 1.645 * sd_x
  RI_upper <- mean_x + 1.645 * sd_x

  results <- boot(x, statistic = bootstrap_RI, R = 10000)

  lower_ci <- boot.ci(results, type = "perc", index = 1)$percent[4:5]
  upper_ci <- boot.ci(results, type = "perc", index = 2)$percent[4:5]
```



```

param_results_fw <- rbind(
  param_results_fw,
  data.frame(
    Variable = v,
    N = n,
    OutliersRemoved = Freshwater_results$Outliers_Removed[Freshwater_results$Variable == v],
    Minimum = min_x,
    Maximum = max_x,
    Mean = mean_x,
    SD = sd_x,
    RI_lower = RI_lower,
    RI_upper = RI_upper,
    CI90_lower_L = lower_ci[1],
    CI90_lower_U = lower_ci[2],
    CI90_upper_L = upper_ci[1],
    CI90_upper_U = upper_ci[2]
  )
)
}

rownames(param_results_fw) <- NULL
param_results_fw

```

##		Variable	N	OutliersRemoved	Minimum	Maximum
## 1		Rectal Temp "C"	20	1	17.00	18.50
## 2		Right IR (F)	21	0	59.00	65.30
## 3		Left IR (F)	21	0	59.00	64.90
## 4		Dorsal IR (F)	21	0	60.40	65.50
## 5		Lac	20	1	1.30	5.40
## 6		HemoCue (g/dl)	21	0	6.70	13.50
## 7		PCV1	20	0	33.00	74.00
## 8		PCV2	20	0	36.00	88.00
## 9		Na (mmol/L)	21	0	132.00	158.00
## 10		Hct (%PCV)	21	0	22.00	56.00
## 11		Hb* (g/dL)	21	0	7.50	19.00
## 12		WBC estimate K/ l	21	0	8.10	22.10
## 13		Absolute Neutrophils K/ l	21	0	1.20	7.00
## 14		Absolute immature neutrophils K/ l	21	0	0.08	0.88
## 15		Absolute Lymphocytes K/ l	21	0	4.80	12.00
## 16		PCV_Avg	21	0	33.00	88.00
##	Mean	SD	RI_lower	RI_upper	CI90_lower_L	CI90_lower_U
## 1	17.5850000	0.4319783	16.87439567	18.2956043	16.73085382	17.0818618
## 2	62.4142857	1.4322310	60.05826568	64.7703058	59.12284831	61.0551585
## 3	62.4952381	1.4667911	60.08236670	64.9081095	59.17538811	61.0602771
## 4	62.5190476	1.4831113	60.07932958	64.9587657	59.50658803	60.9176905
## 5	3.5400000	1.1348081	1.67324064	5.4067594	0.93052406	2.6452454
## 6	10.2000000	2.0697826	6.79520763	13.6047924	5.76934748	8.2640955
## 7	56.9500000	12.3777177	36.58865443	77.3113456	29.31817498	46.8478187
## 8	60.7500000	13.7529901	38.12633128	83.3736687	30.69374879	48.2217415
## 9	147.0476190	5.3616806	138.22765443	155.8675837	133.78836841	142.5996169
## 10	38.6666667	9.8708325	22.42914728	54.9041861	17.99180015	28.9340207
## 11	13.1095238	3.3476715	7.60260412	18.6164435	6.10607427	9.7136043
## 12	14.2190476	3.8086243	7.95386065	20.4842346	6.35126082	10.1967777

```
## 13  3.4523810  1.6995938  0.65654917  6.2482127  0.04178967  1.5445707
## 14  0.3681429  0.2197736  0.00661531  0.7296704 -0.07530511  0.1231582
## 15  8.4047619  2.2384093  4.72257861  12.0869452  3.80184653  6.0223359
## 16  58.9285714  14.2225927  35.53240645  82.3247364  27.85632136  46.1466379
##    CI90_upper_L CI90_upper_U
## 1    17.9249742    18.5762250
## 2    63.7589702    65.5859962
## 3    63.8804883    65.6472833
## 4    63.9180186    65.7327902
## 5     4.8155151     5.8415637
## 6    12.3822956    14.5111488
## 7    71.2885402    81.4240153
## 8    74.9545445    90.3827169
## 9   152.0925839   158.8661119
## 10   48.6661060    59.4850691
## 11   16.4378219    20.1494802
## 12   17.6501622    22.6402119
## 13    4.8226709     7.3769158
## 14    0.5542648     0.8761283
## 15   10.5311301    13.2205233
## 16   73.8708721    89.6084796
```

Finishing off the Gaussian data (n below 20)

```
vars_hist_fw <- Freshwater_results$Variable[
  Freshwater_results$Final_NonNA < 20
]

results_summary_fw <- data.frame()

for (v in vars_hist_fw) {

  x <- na.omit(Freshwater_NoOutliers[[v]])

  hist(x,
    main = paste("Histogram of", v, "(Freshwater - Gaussian)"),
    xlab = v,
    col = "lightblue",
    border = "black")
  cat("\n\\newpage\n")

  n <- length(x)
  mean_val <- mean(x)
  min_val <- min(x)
  max_val <- max(x)

  values_sorted <- paste(sort(x), collapse = ", ")

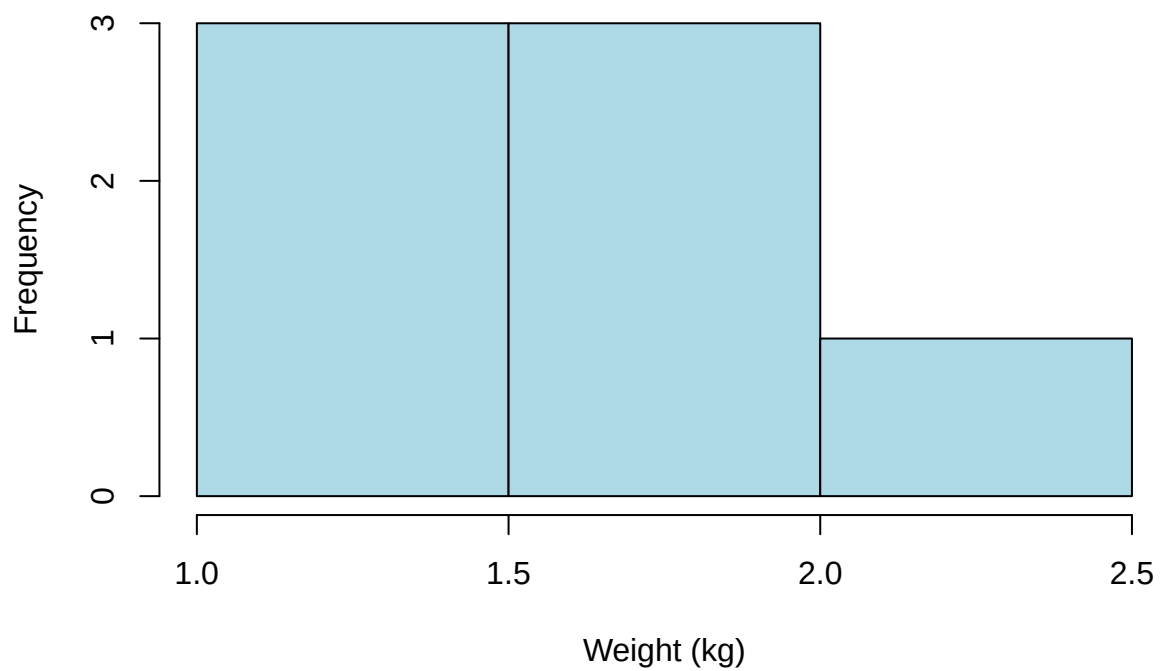
  results_summary_fw <- rbind(
    results_summary_fw,
    data.frame(
      Variable = v,
```

```

N = n,
OutliersRemoved = Freshwater_results$Outliers_Removed[Freshwater_results$Variable == v],
Mean = mean_val,
Minimum = min_val,
Maximum = max_val,
Values = values_sorted
)
)
}

```

Histogram of Weight (kg) (Freshwater – Gaussian)

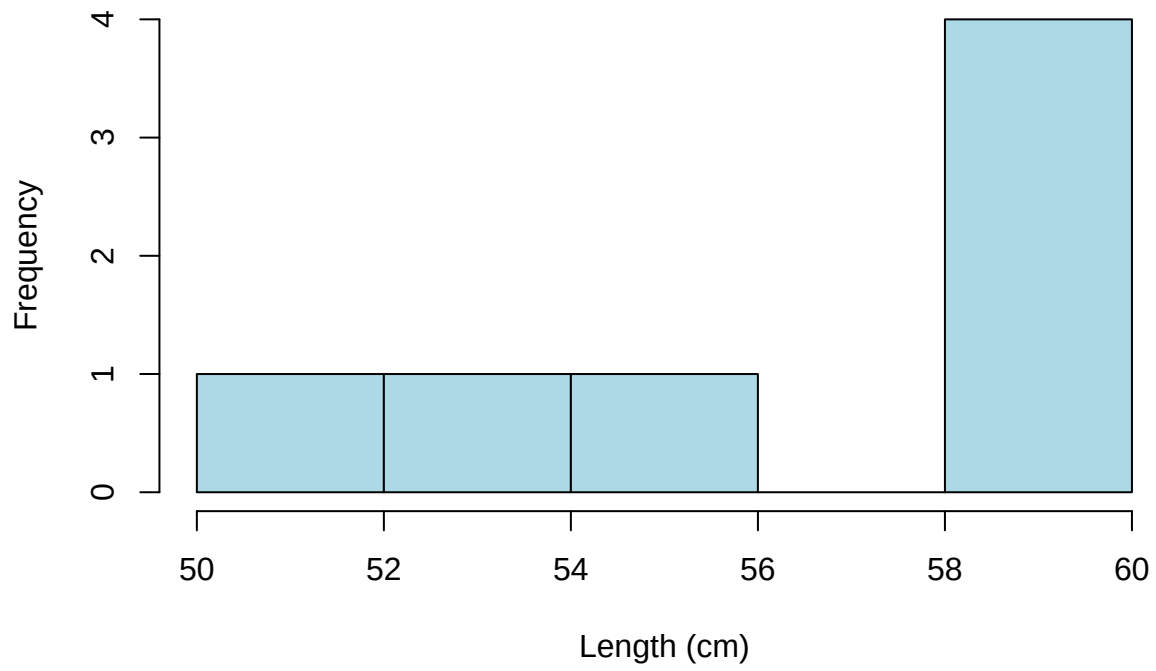


```

##
## \newpage

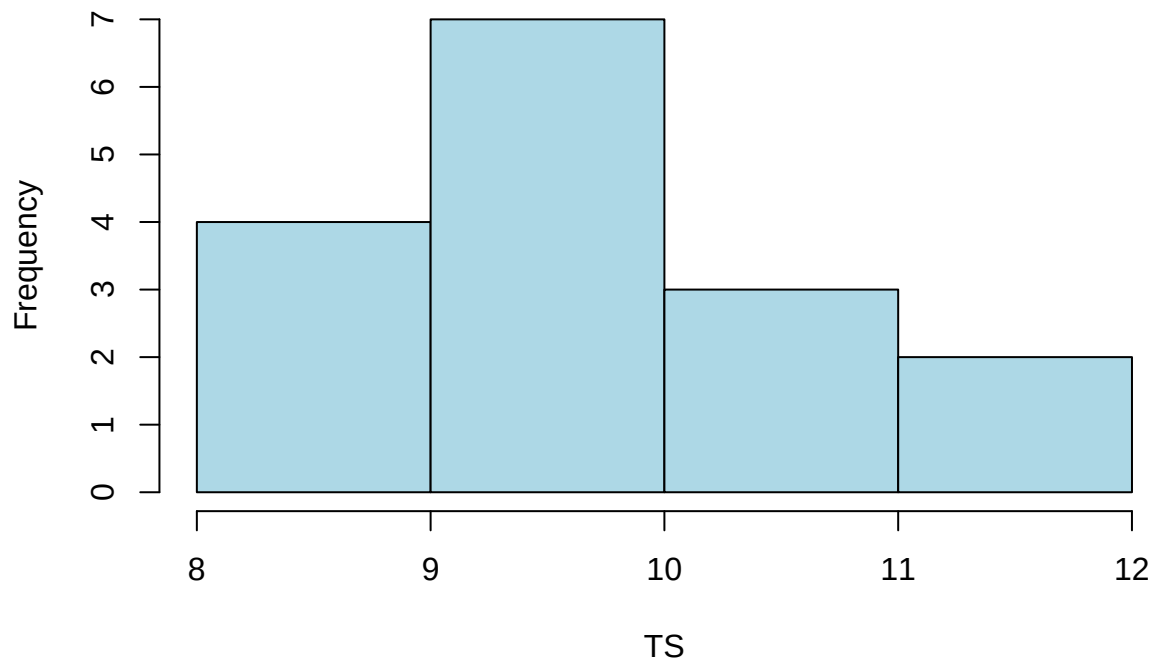
```

Histogram of Length (cm) (Freshwater – Gaussian)



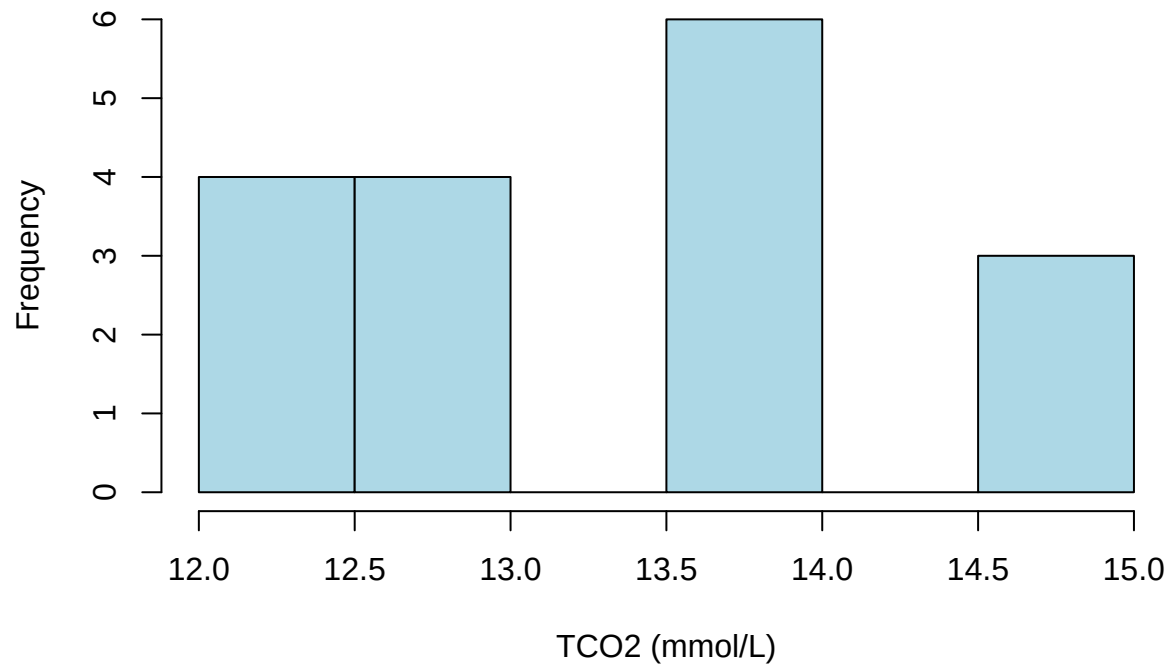
\newpage

Histogram of TS (Freshwater – Gaussian)



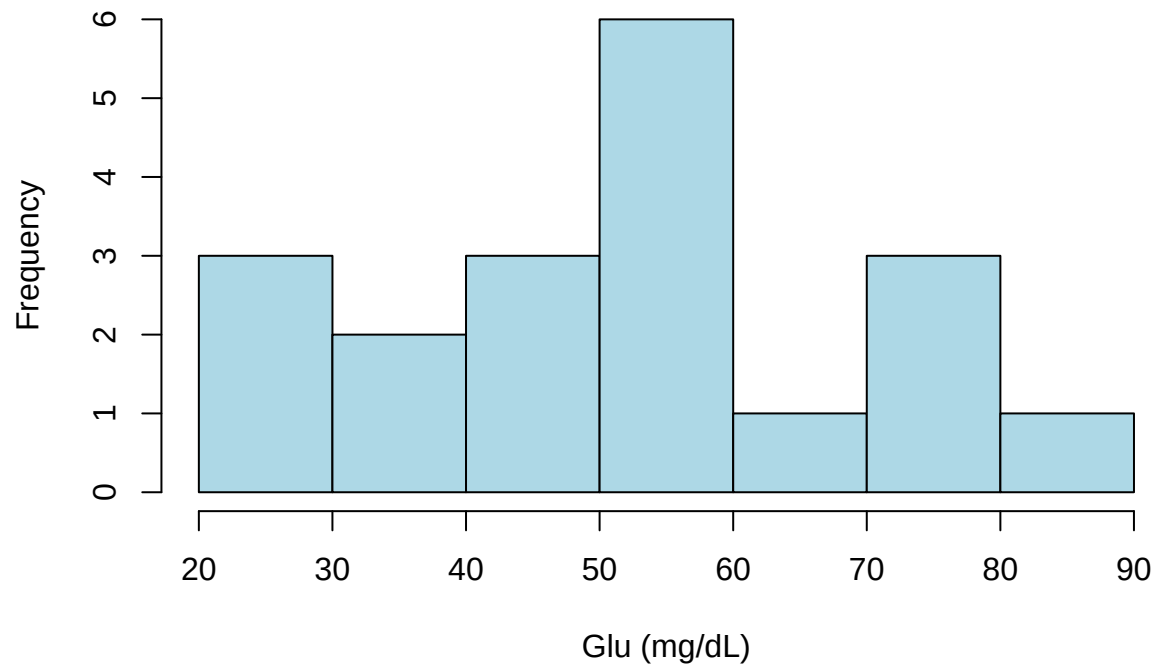
\newpage

Histogram of TCO2 (mmol/L) (Freshwater – Gaussian)



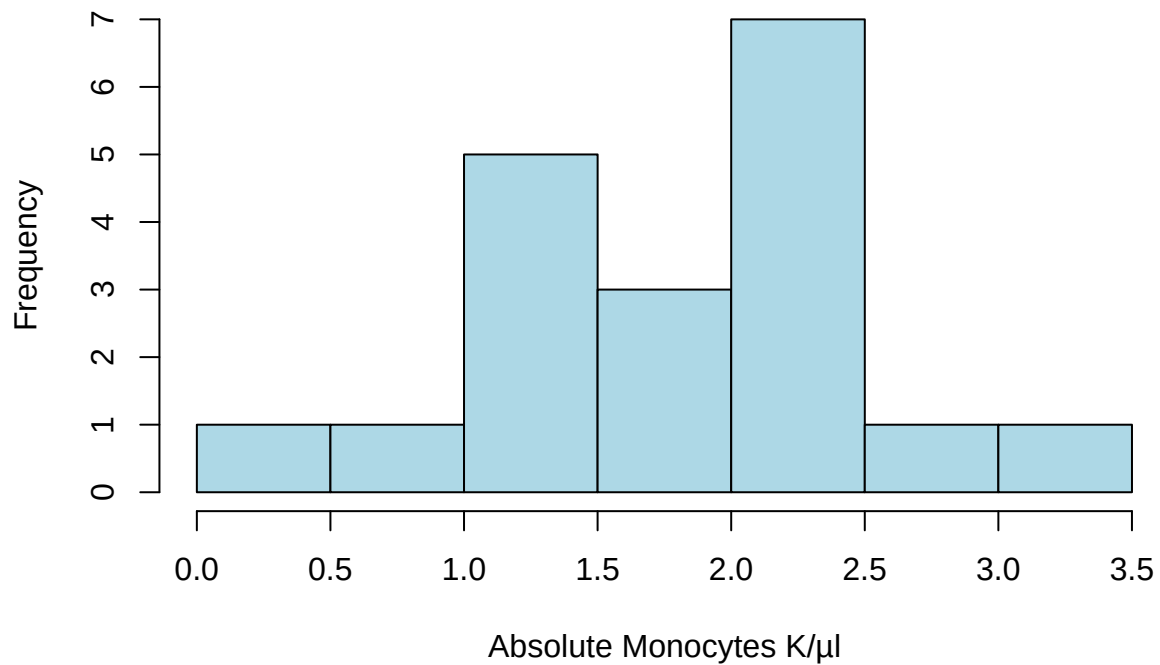
\newpage

Histogram of Glu (mg/dL) (Freshwater – Gaussian)



\newpage

Histogram of Absolute Monocytes K/μl (Freshwater – Gaussian)



```
##
## \newpage
```

```
results_summary_fw
```

```
##          Variable  N OutliersRemoved      Mean Minimum Maximum
## 1      Weight (kg)  7                0  1.692857      1.00      2.2
## 2      Length (cm)  7                0 56.714286     51.50     60.0
## 3          TS     16                2  9.656250      8.20     11.8
## 4      TC02 (mmol/L) 17                3 13.470588     12.00     15.0
## 5          Glu (mg/dL) 19                1 53.000000     22.00     85.0
## 6 Absolute Monocytes K/ l 19                2  1.812105      0.44      3.4
##
##                                     Values
## 1                                     1, 1.5, 1.5, 1.75, 1.9, 2, 2.2
## 2                                     51.5, 53.5, 55, 58.5, 59, 59.5, 60
## 3          8.2, 8.8, 9, 9, 9.1, 9.2, 9.2, 9.2, 9.2, 9.8, 10, 10.2, 10.2, 10.2, 11.4, 11.8
## 4          12, 12, 12, 12, 13, 13, 13, 13, 14, 14, 14, 14, 14, 14, 15, 15, 15
## 5          22, 29, 29, 37, 39, 44, 47, 50, 52, 53, 54, 54, 57, 60, 67, 73, 75, 80, 85
## 6 0.44, 0.59, 1.1, 1.2, 1.3, 1.4, 1.4, 1.7, 1.8, 1.9, 2.1, 2.1, 2.1, 2.2, 2.2, 2.3, 2.5, 2.7, 3.4
```

Freshwater not Gaussian $20 < n < 40$


```

bootstrap_RI_np <- function(data, indices) {

  d <- data[indices]

  c(
    quantile(d, 0.05),
    quantile(d, 0.95)
  )
}

nonnormal_vars_fw <- Freshwater_Normality$Variable[
  !is.na(Freshwater_Normality$Normal) &
  Freshwater_Normality$Normal == "No"
]

vars_np_fw <- nonnormal_vars_fw[
  sapply(nonnormal_vars_fw, function(v)
    sum(!is.na(Freshwater_NoOutliers[[v]])) >= 20 &
    sum(!is.na(Freshwater_NoOutliers[[v]])) < 40
  )
]

nonnormal_results_fw <- data.frame()

for (v in vars_np_fw) {

  x <- na.omit(Freshwater_NoOutliers[[v]])

  n <- length(x)
  min_x <- min(x)
  max_x <- max(x)
  med_x <- median(x)

  RI_lower <- quantile(x, 0.05)
  RI_upper <- quantile(x, 0.95)

  results <- boot(x, statistic = bootstrap_RI_np, R = 10000)

  lower_ci <- boot.ci(results, type = "perc", index = 1)$percent[4:5]
  upper_ci <- boot.ci(results, type = "perc", index = 2)$percent[4:5]

  nonnormal_results_fw <- rbind(
    nonnormal_results_fw,
    data.frame(
      Variable = v,
      N = n,
      Minimum = min_x,
      Maximum = max_x,
      Median = med_x,
      RI_lower = RI_lower,
      RI_upper = RI_upper,
      CI90_lower_L = lower_ci[1],
      CI90_lower_U = lower_ci[2],

```

```

    CI90_upper_L = upper_ci[1],
    CI90_upper_U = upper_ci[2]
  )
}

rownames(nonnormal_results_fw) <- NULL
nonnormal_results_fw

```

```

##              Variable  N Minimum Maximum Median RI_lower RI_upper
## 1      Heart Rate (/min) 21   12.00   48.0  24.00   16.00   44.00
## 2              iCa (mmol/L) 21    1.22    1.8   1.64    1.28    1.79
## 3 #immature RBC/100 mature RBC 21    2.00   11.0   4.00    2.00    9.00
##  CI90_lower_L CI90_lower_U CI90_upper_L CI90_upper_U
## 1      12.00      20.00      32.00      48.0
## 2       1.22       1.59       1.69       1.8
## 3       2.00       2.00       6.00      11.0

```

Freshwater not Gaussian (n below 20)

```

vars_small_np_fw <- nonnormal_vars_fw[
  sapply(nonnormal_vars_fw, function(v)
    sum(!is.na(Freshwater[[v]])) < 20
  )
]

results_summary_small_np_fw <- data.frame()

for (v in vars_small_np_fw) {
  x <- na.omit(Freshwater[[v]])

  hist(x,
    main = paste("Histogram of", v, "(Freshwater - Not Gaussian)",
    xlab = v,
    col = "lightblue",
    border = "black")

  cat("\n\\newpage\n")

  n <- length(x)
  median_val <- median(x)
  min_val <- min(x)
  max_val <- max(x)

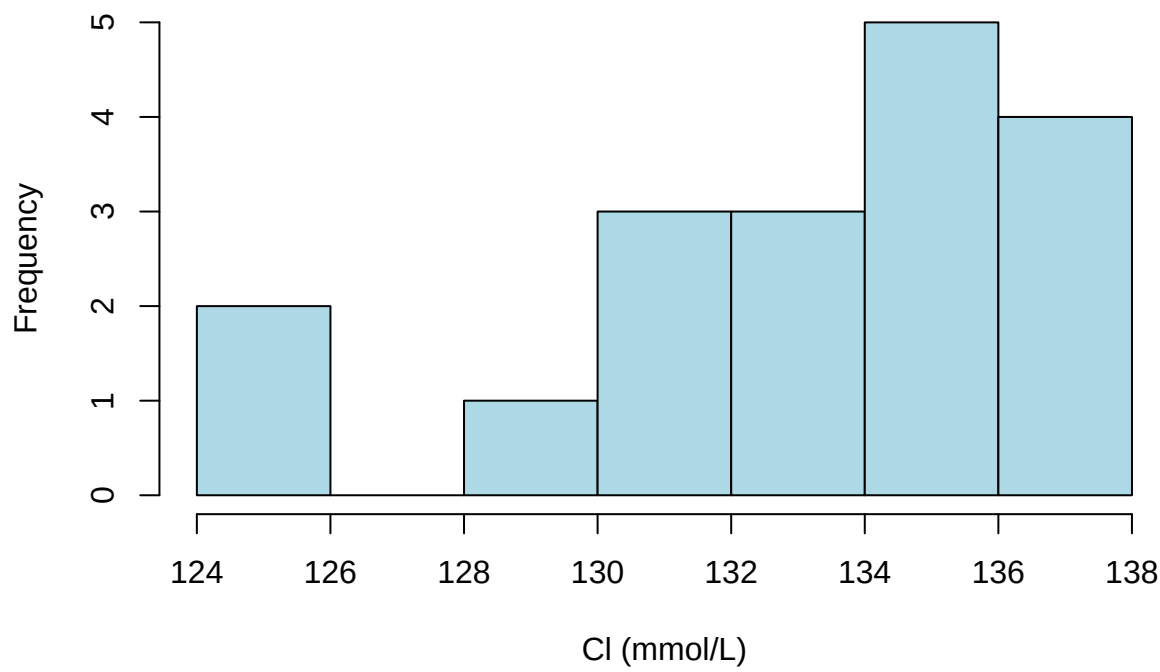
  values_sorted <- paste(sort(x), collapse = ", ")

  results_summary_small_np_fw <- rbind(
    results_summary_small_np_fw,
    data.frame(
      Variable = v,

```

```
N = n,  
Median = median_val,  
Minimum = min_val,  
Maximum = max_val,  
Values = values_sorted  
)  
)  
}
```

Histogram of Cl (mmol/L) (Freshwater – Not Gaussian)



```
##  
## \newpage
```

```
results_summary_small_np_fw
```

```
##      Variable  N Median Minimum Maximum  
## 1 Cl (mmol/L) 18   134    125     138  
##  
## 1 125, 125, 130, 131, 132, 132, 133, 133, 133, 135, 135, 136, 136, 136, 137, 137, 137, 138
```

Freshwater Results

param_results_fw

##	Variable		N	OutliersRemoved	Minimum	Maximum
## 1	Rectal Temp "C"	20		1	17.00	18.50
## 2	Right IR (F)	21		0	59.00	65.30
## 3	Left IR (F)	21		0	59.00	64.90
## 4	Dorsal IR (F)	21		0	60.40	65.50
## 5	Lac	20		1	1.30	5.40
## 6	HemoCue (g/dl)	21		0	6.70	13.50
## 7	PCV1	20		0	33.00	74.00
## 8	PCV2	20		0	36.00	88.00
## 9	Na (mmol/L)	21		0	132.00	158.00
## 10	Hct (%PCV)	21		0	22.00	56.00
## 11	Hb* (g/dL)	21		0	7.50	19.00
## 12	WBC estimate K/ l	21		0	8.10	22.10
## 13	Absolute Neutrophils K/ l	21		0	1.20	7.00
## 14	Absolute immature neutrophils K/ l	21		0	0.08	0.88
## 15	Absolute Lymphocytes K/ l	21		0	4.80	12.00
## 16	PCV_Avg	21		0	33.00	88.00
##	Mean	SD	RI_lower	RI_upper	CI90_lower_L	CI90_lower_U
## 1	17.5850000	0.4319783	16.87439567	18.2956043	16.73085382	17.0818618
## 2	62.4142857	1.4322310	60.05826568	64.7703058	59.12284831	61.0551585
## 3	62.4952381	1.4667911	60.08236670	64.9081095	59.17538811	61.0602771
## 4	62.5190476	1.4831113	60.07932958	64.9587657	59.50658803	60.9176905
## 5	3.5400000	1.1348081	1.67324064	5.4067594	0.93052406	2.6452454
## 6	10.2000000	2.0697826	6.79520763	13.6047924	5.76934748	8.2640955
## 7	56.9500000	12.3777177	36.58865443	77.3113456	29.31817498	46.8478187
## 8	60.7500000	13.7529901	38.12633128	83.3736687	30.69374879	48.2217415
## 9	147.0476190	5.3616806	138.22765443	155.8675837	133.78836841	142.5996169
## 10	38.6666667	9.8708325	22.42914728	54.9041861	17.99180015	28.9340207
## 11	13.1095238	3.3476715	7.60260412	18.6164435	6.10607427	9.7136043
## 12	14.2190476	3.8086243	7.95386065	20.4842346	6.35126082	10.1967777
## 13	3.4523810	1.6995938	0.65654917	6.2482127	0.04178967	1.5445707
## 14	0.3681429	0.2197736	0.00661531	0.7296704	-0.07530511	0.1231582
## 15	8.4047619	2.2384093	4.72257861	12.0869452	3.80184653	6.0223359
## 16	58.9285714	14.2225927	35.53240645	82.3247364	27.85632136	46.1466379
##	CI90_upper_L	CI90_upper_U				
## 1	17.9249742	18.5762250				
## 2	63.7589702	65.5859962				
## 3	63.8804883	65.6472833				
## 4	63.9180186	65.7327902				
## 5	4.8155151	5.8415637				
## 6	12.3822956	14.5111488				
## 7	71.2885402	81.4240153				
## 8	74.9545445	90.3827169				
## 9	152.0925839	158.8661119				
## 10	48.6661060	59.4850691				
## 11	16.4378219	20.1494802				
## 12	17.6501622	22.6402119				
## 13	4.8226709	7.3769158				
## 14	0.5542648	0.8761283				
## 15	10.5311301	13.2205233				
## 16	73.8708721	89.6084796				

```
results_summary_fw
```

```
##           Variable  N OutliersRemoved      Mean Minimum Maximum
## 1      Weight (kg)  7              0 1.692857      1.00      2.2
## 2      Length (cm)  7              0 56.714286     51.50     60.0
## 3              TS 16              2  9.656250      8.20     11.8
## 4      TC02 (mmol/L) 17            3 13.470588     12.00     15.0
## 5          Glu (mg/dL) 19            1 53.000000     22.00     85.0
## 6 Absolute Monocytes K/ l 19          2  1.812105      0.44      3.4
##
##                                                     Values
## 1                                                     1, 1.5, 1.5, 1.75, 1.9, 2, 2.2
## 2                                                     51.5, 53.5, 55, 58.5, 59, 59.5, 60
## 3          8.2, 8.8, 9, 9, 9.1, 9.2, 9.2, 9.2, 9.2, 9.8, 10, 10.2, 10.2, 10.2, 11.4, 11.8
## 4              12, 12, 12, 12, 13, 13, 13, 13, 14, 14, 14, 14, 14, 14, 15, 15, 15
## 5          22, 29, 29, 37, 39, 44, 47, 50, 52, 53, 54, 54, 57, 60, 67, 73, 75, 80, 85
## 6 0.44, 0.59, 1.1, 1.2, 1.3, 1.4, 1.4, 1.7, 1.8, 1.9, 2.1, 2.1, 2.1, 2.2, 2.2, 2.3, 2.5, 2.7, 3.4
```

```
nonnormal_results_fw
```

```
##           Variable  N Minimum Maximum Median RI_lower RI_upper
## 1      Heart Rate (/min) 21    12.00    48.0  24.00    16.00    44.00
## 2              iCa (mmol/L) 21     1.22     1.8   1.64     1.28     1.79
## 3 #immature RBC/100 mature RBC 21     2.00    11.0   4.00     2.00     9.00
##  CI90_lower_L CI90_lower_U CI90_upper_L CI90_upper_U
## 1      12.00      20.00      32.00      48.0
## 2      1.22      1.59      1.69      1.8
## 3      2.00      2.00      6.00      11.0
```

```
results_summary_small_np_fw
```

```
##           Variable  N Median Minimum Maximum
## 1 Cl (mmol/L) 18    134     125     138
##
##                                                     Values
## 1 125, 125, 130, 131, 132, 132, 133, 133, 133, 135, 135, 136, 136, 136, 137, 137, 137, 138
```

```
Freshwater_Normality$Variable[is.na(Freshwater_Normality$Normal)]
```

```
## [1] "K (mmol/L)"          "BUN (mg/dL)"
## [3] "Crea (mg/dL)"        "Absolute Eosinophils K/ l"
## [5] "Absolute Basophils K/ l"
```

T-Tests and BH Correction

```
p_values <- numeric(length(vars))
mean_salt <- numeric(length(vars))
mean_fresh <- numeric(length(vars))
difference <- numeric(length(vars))
se_diff <- numeric(length(vars))
```

```

t_ratio <- numeric(length(vars))
df_val <- numeric(length(vars))

names(p_values) <- vars
names(mean_salt) <- vars
names(mean_fresh) <- vars
names(difference) <- vars
names(se_diff) <- vars
names(t_ratio) <- vars
names(df_val) <- vars

for (v in vars) {

  x <- na.omit(Saltwater[[v]])
  y <- na.omit(Freshwater[[v]])

  if (length(x) > 2 && length(y) > 2) {

    test <- t.test(x, y)

    p_values[v] <- test$p.value
    mean_salt[v] <- mean(x)
    mean_fresh[v] <- mean(y)
    difference[v] <- mean(x) - mean(y)
    se_diff[v] <- test$stderr
    t_ratio[v] <- test$statistic
    df_val[v] <- test$parameter

  } else {

    p_values[v] <- NA
    mean_salt[v] <- NA
    mean_fresh[v] <- NA
    difference[v] <- NA
    se_diff[v] <- NA
    t_ratio[v] <- NA
    df_val[v] <- NA

  }
}

q_values <- p.adjust(p_values, method = "BH")

comparison_results <- data.frame(
  Variable = vars,
  Mean_Saltwater = round(mean_salt, 3),
  Mean_Freshwater = round(mean_fresh, 3),
  Difference = round(difference, 3),
  SE_Difference = round(se_diff, 3),
  T_ratio = round(t_ratio, 3),
  DF = round(df_val, 2),
  P_value = signif(p_values, 3),
  Q_value = signif(q_values, 3),

```

```

Significant = ifelse(q_values < 0.05, "Yes", "No")
)

comparison_results

```

##	Variable
## Heart Rate (/min)	Heart Rate (/min)
## Rectal Temp "C"	Rectal Temp "C"
## Right IR (F)	Right IR (F)
## Left IR (F)	Left IR (F)
## Dorsal IR (F)	Dorsal IR (F)
## Weight (kg)	Weight (kg)
## Length (cm)	Length (cm)
## Lac	Lac
## HemoCue (g/dl)	HemoCue (g/dl)
## PCV1	PCV1
## PCV2	PCV2
## TS	TS
## Na (mmol/L)	Na (mmol/L)
## K (mmol/L)	K (mmol/L)
## Cl (mmol/L)	Cl (mmol/L)
## iCa (mmol/L)	iCa (mmol/L)
## TC02 (mmol/L)	TC02 (mmol/L)
## Glu (mg/dL)	Glu (mg/dL)
## Hct (%PCV)	Hct (%PCV)
## Hb* (g/dL)	Hb* (g/dL)
## BUN (mg/dL)	BUN (mg/dL)
## Crea (mg/dL)	Crea (mg/dL)
## WBC estimate K/ l	WBC estimate K/ l
## Absolute Neutrophils K/ l	Absolute Neutrophils K/ l
## Absolute immature neutrophils K/ l	Absolute immature neutrophils K/ l
## Absolute Lymphocytes K/ l	Absolute Lymphocytes K/ l
## Absolute Monocytes K/ l	Absolute Monocytes K/ l
## Absolute Eosinophils K/ l	Absolute Eosinophils K/ l
## Absolute Basophils K/ l	Absolute Basophils K/ l
## #immature RBC/100 mature RBC	#immature RBC/100 mature RBC
## PCV_Avg	PCV_Avg
##	Mean_Saltwater Mean_Freshwater Difference
## Heart Rate (/min)	17.520 26.286 -8.766
## Rectal Temp "C"	12.180 17.519 -5.339
## Right IR (F)	54.892 62.414 -7.522
## Left IR (F)	54.644 62.495 -7.851
## Dorsal IR (F)	54.132 62.519 -8.387
## Weight (kg)	2.839 1.693 1.146
## Length (cm)	63.840 56.714 7.126
## Lac	2.976 3.743 -0.767
## HemoCue (g/dl)	9.740 10.200 -0.460
## PCV1	42.920 56.950 -14.030
## PCV2	43.458 60.750 -17.292
## TS	7.848 9.350 -1.502
## Na (mmol/L)	151.400 147.048 4.352
## K (mmol/L)	NA NA NA
## Cl (mmol/L)	137.500 133.389 4.111

## iCa (mmol/L)	1.518	1.611	-0.093
## TC02 (mmol/L)	9.160	13.550	-4.390
## Glu (mg/dL)	65.440	56.600	8.840
## Hct (%PCV)	27.360	38.667	-11.307
## Hb* (g/dL)	9.368	13.110	-3.742
## BUN (mg/dL)	NA	NA	NA
## Crea (mg/dL)	NA	NA	NA
## WBC estimate K/ l	10.847	14.219	-3.372
## Absolute Neutrophils K/ l	3.487	3.452	0.034
## Absolute immature neutrophils K/ l	0.320	0.368	-0.048
## Absolute Lymphocytes K/ l	6.367	8.405	-2.038
## Absolute Monocytes K/ l	0.699	1.992	-1.293
## Absolute Eosinophils K/ l	0.000	0.000	0.000
## Absolute Basophils K/ l	0.000	0.000	0.000
## #immature RBC/100 mature RBC	28.875	4.524	24.351
## PCV_Avg	43.200	58.929	-15.729
##	SE_Difference	T_ratio	DF P_value
## Heart Rate (/min)	2.783	-3.149	42.23 3.00e-03
## Rectal Temp "C"	0.221	-24.163	38.28 8.83e-25
## Right IR (F)	0.736	-10.215	33.75 7.33e-12
## Left IR (F)	0.785	-10.000	32.95 1.65e-11
## Dorsal IR (F)	0.672	-12.474	36.66 9.32e-15
## Weight (kg)	0.229	4.994	22.52 5.01e-05
## Length (cm)	1.521	4.685	11.88 5.42e-04
## Lac	0.398	-1.924	39.33 6.16e-02
## HemoCue (g/dl)	0.537	-0.856	35.00 3.98e-01
## PCV1	3.056	-4.591	27.21 9.00e-05
## PCV2	3.363	-5.142	26.33 2.23e-05
## TS	0.439	-3.418	40.65 1.44e-03
## Na (mmol/L)	1.340	3.248	31.83 2.74e-03
## K (mmol/L)	NA	NA	NA NA
## Cl (mmol/L)	1.181	3.481	18.22 2.63e-03
## iCa (mmol/L)	0.043	-2.141	43.33 3.80e-02
## TC02 (mmol/L)	0.568	-7.735	42.90 1.15e-09
## Glu (mg/dL)	5.465	1.617	22.02 1.20e-01
## Hct (%PCV)	2.402	-4.707	29.47 5.52e-05
## Hb* (g/dL)	0.822	-4.554	30.24 8.06e-05
## BUN (mg/dL)	NA	NA	NA NA
## Crea (mg/dL)	NA	NA	NA NA
## WBC estimate K/ l	1.276	-2.642	30.59 1.29e-02
## Absolute Neutrophils K/ l	0.643	0.053	26.80 9.58e-01
## Absolute immature neutrophils K/ l	0.079	-0.607	28.25 5.48e-01
## Absolute Lymphocytes K/ l	0.667	-3.057	33.63 4.36e-03
## Absolute Monocytes K/ l	0.218	-5.941	28.74 1.94e-06
## Absolute Eosinophils K/ l	0.000	NaN	NaN NaN
## Absolute Basophils K/ l	0.000	NaN	NaN NaN
## #immature RBC/100 mature RBC	2.380	10.231	25.92 1.36e-10
## PCV_Avg	3.364	-4.675	26.92 7.33e-05
##	Q_value	Significant	
## Heart Rate (/min)	4.34e-03	Yes	
## Rectal Temp "C"	2.30e-23	Yes	
## Right IR (F)	6.36e-11	Yes	
## Left IR (F)	1.07e-10	Yes	
## Dorsal IR (F)	1.21e-13	Yes	

## Weight (kg)	1.44e-04	Yes
## Length (cm)	1.01e-03	Yes
## Lac	7.27e-02	No
## HemoCue (g/dl)	4.31e-01	No
## PCV1	1.80e-04	Yes
## PCV2	7.25e-05	Yes
## TS	2.50e-03	Yes
## Na (mmol/L)	4.19e-03	Yes
## K (mmol/L)	NA	<NA>
## Cl (mmol/L)	4.19e-03	Yes
## iCa (mmol/L)	4.70e-02	Yes
## TC02 (mmol/L)	4.97e-09	Yes
## Glu (mg/dL)	1.36e-01	No
## Hct (%PCV)	1.44e-04	Yes
## Hb* (g/dL)	1.75e-04	Yes
## BUN (mg/dL)	NA	<NA>
## Crea (mg/dL)	NA	<NA>
## WBC estimate K/ l	1.67e-02	Yes
## Absolute Neutrophils K/ l	9.58e-01	No
## Absolute immature neutrophils K/ l	5.70e-01	No
## Absolute Lymphocytes K/ l	5.97e-03	Yes
## Absolute Monocytes K/ l	7.21e-06	Yes
## Absolute Eosinophils K/ l	NaN	<NA>
## Absolute Basophils K/ l	NaN	<NA>
## #immature RBC/100 mature RBC	7.07e-10	Yes
## PCV_Avg	1.73e-04	Yes

Exporting Results to CSVs